

CE 310 — Week 11 Assignment

Week: 11 | **Topic:** Hypothesis Testing — t-Tests and ANOVA **Points:** 60 pts total (3 problems) **Due:** Wednesday of Week 12 by 9:00 AM — **submit to D2L** —

Overview

The lab established the core mechanics of hypothesis testing applied to the DOE Medium Office dataset. This assignment asks you to extend those tests to new subsets and scenarios, reason carefully about effect size vs. statistical significance, and apply hypothesis testing to a track-specific engineering context.

You will need: your completed lab notebook (Week11_Lab.ipynb), the DOE dataset (Week5_DOE_MedOffice_ZoneTemp_2023.csv), and a Python environment (Google Colab or local).

How to Submit

Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Run all cells	Runtime → Run all — wait until every cell shows a checkmark, no spinners
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Print lines	Do not edit or delete the <code>print(f"ANSWER_A1 = ...")</code> lines
File name	<code>lastname_firstname_weekN.ipynb</code> e.g. <code>smith_jane_week11.ipynb</code>
Submission	Upload the .ipynb file to D2L Dropbox before the deadline

Problem 1 — Hypothesis Testing Extensions (30 pts)

Do not repeat the lab computations. Use the lab results as context; this problem requires new tests on subsets and scenarios not covered in the lab.

P1a (8 pts) — Nighttime vs. Daytime Cooling

Business-hours occupancy is one way to partition the data, but engineers often think in terms of **nighttime** (economizer potential, free cooling) vs. **daytime** (peak load hours).

Define two groups: - **Nighttime:** Hour < 6 or Hour >= 22 (10 pm – 5 am, 8 hours per day) - **Daytime:** all remaining hours (6 ≤ Hour < 22, 16 hours per day)

```
night = df[(df['Hour'] < 6) | (df['Hour'] >= 22)][['Cooling_kWh']]
day    = df[(df['Hour'] >= 6) & (df['Hour'] < 22)][['Cooling_kWh']]
```

Run a two-sample t-test and compute Cohen's d. Report: - Mean cooling for each group (kWh/hr) - t-statistic and p-value - Cohen's d and its conventional classification (small/medium/large) - Your conclusion: reject or fail to reject H₀ at α = 0.05?

Compare Cohen's d here to the BH-group d (d = 1.30 from the lab). Which partition — nighttime/daytime or business-hours/non-business-hours — produces the larger effect size? Explain in one sentence why one partition might capture the occupancy signal more cleanly.

P1b (8 pts) — Summer Zone Temperature Compliance

A common cooling setpoint for commercial office buildings is **75°F**. The lab showed the annual mean zone temperature is 73.55°F — below the setpoint overall. But summer is the critical season for overheating risk.

Filter the dataset to **summer, occupied hours only** (Season == 'Summer' and BH == 1).

Run a one-sample t-test: - **H₀:** zone_summer_occupied = 75°F - **H_a:** zone_summer_occupied > 75°F

Report the subset mean, its standard deviation, the t-statistic, p-value, and your conclusion at α = 0.05.

Then compute what fraction of these summer occupied hours have Zone_Temp_F > 75°F. Does the t-test result (reject or fail to reject) seem consistent with the raw fraction above? In 2–3 sentences, explain how a very small standard deviation combined with a large sample can produce a statistically significant t-test even when the mean deviation from the setpoint is practically negligible — and what this implies about using means vs. distributions for thermostat compliance checks.

P1c (7 pts) — Month-Level Post-Hoc Analysis

The lab ANOVA by month (B4) showed $F = 331.6$, $p = 0$. Run a **Tukey HSD** post-hoc test on monthly cooling means:

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd
tukey_month = pairwise_tukeyhsd(df['Cooling_kWh'], df['Month'], alpha=0.05)
print(tukey_month)
```

From the output, identify: - The **month pair with the largest mean difference** in cooling. Report the two months and the mean difference (kWh/hr). - How many month pairs are **NOT significantly different** at $\alpha = 0.05$ (**reject = False**)? Choose **one** such non-significant pair and explain in 2–3 sentences why those two months have similar cooling loads despite being different calendar months. (Hint: look at the monthly mean cooling table from B4 and find pairs with nearly identical means.)

P1d (7 pts) — Effect Size vs. Sample Size

You have three t-test results from the lab:

Comparison	t-stat	Cohen's d
Cooling: BH=1 vs BH=0	55.68	1.30
Cooling: Weekday vs Weekend	16.01	0.38
Outdoor Temp: BH=1 vs BH=0	26.76	0.63

The BH cooling comparison has both the largest t-statistic AND the largest Cohen's d. The Weekday/Weekend comparison has the smallest t-stat AND the smallest d.

(a) Imagine you had only 100 hours of data (randomly sampled) instead of 8,760. Without re-running Python, predict: which of the three tests would most likely **fail** to reach statistical significance ($p > 0.05$) at $n = 100$? Justify your answer using the relationship between Cohen's d, sample size, and statistical power.

(b) The Outdoor Temp comparison ($d = 0.63$) has a *larger* d than the Weekday/Weekend comparison ($d = 0.38$) yet a *higher* t-stat (26.76 vs. 16.01). Is this consistent or inconsistent with your understanding of what controls the t-statistic? Explain in 2 sentences.

(c) An engineering colleague says: "With 8,760 observations, every t-test will give $p < 0.05$ — so hypothesis testing is useless at this scale." Write a 2-sentence rebuttal that distinguishes what p-values and effect sizes tell you, and when each matters for engineering decisions.

Problem 2 — Track Application (18 pts)

Choose one track. CE students complete Track A. ArcE students complete Track B.

Track A — CE: Equipment Sizing Under Uncertainty (18 pts)

A mechanical engineer is deciding whether to size cooling equipment separately for occupied vs. unoccupied operation, rather than using a single “average” design point.

(A2-a, 6 pts) Run a one-sample t-test for **occupied hours only** ($BH = 1$) against a benchmark of **35 kWh/hr** (the expected average cooling load for this climate zone and building type during occupied periods, per a published ASHRAE guidelines reference):

```
t_occ, p_occ = stats.ttest_1samp(occ, 35)
```

Report the sample mean, t-statistic, p-value, and your conclusion at $\alpha = 0.05$. Does this building’s occupied cooling load significantly exceed the 35 kWh/hr benchmark?

(A2-b, 6 pts) Now test **unoccupied hours only** ($BH = 0$) against a benchmark of **12 kWh/hr** (the expected average for unoccupied periods):

```
t_unocc, p_unocc = stats.ttest_1samp(unocc, 12)
```

Report the sample mean, t-statistic, p-value, and your conclusion.

Compare the two conclusions: the building exceeds the occupied benchmark by how many kWh/hr? The unoccupied load: is it above or below the benchmark, and by how much? In one sentence, state which operating mode presents the larger *relative* deviation from the benchmark.

(A2-c, 6 pts) An engineer sizes a chiller using the **95th percentile** of cooling demand, not the mean, to ensure the system handles peak loads.

Compute the 95th percentile of Cooling_kWh for occupied hours using:

```
p95_occ = occ.quantile(0.95)
```

Then test whether the **mean** occupied cooling differs significantly from this 95th-percentile value using a one-sample t-test. Report the t-statistic and p-value.

In 2–3 sentences, explain why an engineer should use the 95th-percentile value — not the mean — as the design point for chiller sizing, and what role hypothesis testing plays (if any) in this decision.

Track B — ArcE: ASHRAE 55 Thermal Comfort by Season (18 pts)

ASHRAE 55-2020 specifies operative temperature ranges for thermal comfort. For a mechanically cooled office in a hot-dry climate, the standard upper limit for 80% acceptability in cooling mode is approximately **79°F** operative temperature during occupied hours.

(B2-a, 6 pts) Filter to **all occupied hours** ($BH = 1$). Run a one-sample t-test against a cooling setpoint of **75°F**: - **H**: `_zone_occupied = 75°F` - **H**: `_zone_occupied > 75°F`

Report the occupied mean `Zone_Temp_F`, its standard deviation, the t-statistic, p-value, and conclusion at $\alpha = 0.05$.

Then compute the fraction of occupied hours where `Zone_Temp_F > 75°F`. The t-test may indicate the mean is significantly different from 75°F — but what does the fraction above 75°F tell you about how often the building actually exceeds the setpoint? In 2 sentences, reconcile the t-test conclusion with the practical implication for occupant comfort.

(B2-b, 6 pts) Split occupied hours by season and compute mean and standard deviation of `Zone_Temp_F` for each season:

```
occ_df = df[df['BH'] == 1]
occ_df.groupby('Season')['Zone_Temp_F'].agg(['mean', 'std', 'count']).round(3)
```

Run a one-way ANOVA on zone temperature for **occupied hours only**, across the four seasons. Report the F-statistic and p-value.

Which season has the warmest occupied zone temperature? Is this consistent with the HVAC system working harder in summer to cool an already-warm building, or does it suggest the system is struggling to maintain setpoint?

(B2-c, 6 pts) Compare Summer vs. Winter occupied zone temperatures using a two-sample t-test. Report t-statistic, p-value, and Cohen's d.

Using the Cohen's d benchmarks ($d < 0.2$ negligible · $0.2-0.5$ small · $0.5-0.8$ medium · > 0.8 large), classify the effect. In 2–3 sentences, explain what this effect size implies for an architect designing separate HVAC setpoint schedules for summer and winter — specifically, does the magnitude of the seasonal zone-temperature difference justify different summer vs. winter setpoints in the building controls system?

Problem 3 — Written Reflection (12 pts)

Write a **100–150 word** reflection addressing the following prompt:

Hypothesis testing gave you a formal way to confirm what the descriptive statistics and regression analysis in Weeks 9–10 already suggested. Describe **one situation** where hypothesis testing would

change your engineering conclusion compared to looking at the raw means alone, and **one situation** where looking at the raw means is sufficient and hypothesis testing adds little practical value. Support each with a **specific number** from this week's lab (a t-statistic, p-value, Cohen's d, or mean difference).

Grading criteria (12 pts total): - First scenario: specific, plausible, grounded in a lab number (4 pts) - Second scenario: specific, plausible, grounded in a lab number (4 pts) - Writing quality: concise, within 100–150 words, evidence used appropriately (4 pts)

Submission Checklist

- ☐ P1a: Python output shown; t-stat, p-value, Cohen's d reported; comparison to BH-group d
- ☐ P1b: Subset filtered; t-test on summer occupied zone temp vs. 75°F; fraction > 75°F computed; 2–3 sentence explanation
- ☐ P1c: Tukey HSD on months; largest-difference pair identified; non-significant pairs listed and explained
- ☐ P1d: (a) prediction with justification; (b) explanation of t vs. d; (c) rebuttal of colleague's claim
- ☐ P2 (Track A or B): all three sub-parts with code output and written responses
- ☐ P3: reflection, 100–150 words, two scenarios, two quantitative observations